

# CS101 - Algorithms & Programming I

Spring 2026 - Lab 2

Due: Week of February 16, 2026

Remember the **honor code** for your programming assignments.

For all labs, your solutions must conform to the CS101 style **guidelines**!

All data and results should be stored in variables (or constants where appropriate) with meaningful names.

The objective of this lab is to write basic Java programs that take inputs from the user and generate respective outputs on the console/terminal. The outputs are expected to have a certain formatting to achieve user-friendliness. As always, this process will include program design & debugging. Remember that analyzing your problems and designing them on a piece of paper *before* starting implementation/coding is always a best practice. Inputs of the sample runs are shown in **blue**.

## 0. Setup Workspace

Start VSC and open the previously created folder named `labs`. Now, under the `labs` folder, create a new folder named `lab2`. In this lab, you are to have three Java classes/files (under `labs/lab2` folder) as described below. Two more Java files containing the revisions should go under this folder as well. We expect you to submit a total of 5 files, including the revision, **without compressing** them. Do *not* upload other/previous lab solutions in your submission.

## 1. User I/O Calculations

Create a new/empty file of your own under the `lab2` folder named `Lab02_Q1.java` with a class with the same name that takes two **double** inputs from the user. Using the base edge length and slant height, determine the vertical height, volume and total surface area. The user inputs are shown in blue below

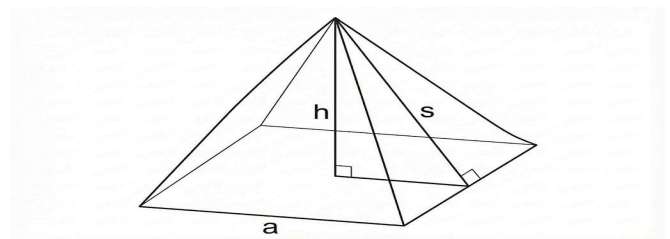
User inputs are:

- 1-) The **base edge length (a)** of the pyramid.
- 2-) The **slant height (s)** of the pyramid.

$$V = \frac{1}{3}a^2h$$

$$h = \sqrt{s^2 - \left(\frac{a}{2}\right)^2}$$

$$A = a^2 + 2as$$



and outputs the results as below:

```
Enter the base edge length: 10.0
Enter the slant height: 13.0
The height of the pyramid is: 12.0
The volume of the pyramid is: 400.0
The total surface area is: 360.0
```

Make sure that the type of values in these expressions is **coded exactly as provided**. Also, make sure that the output is **formatted exactly as shown**.

## 2. Formatted Output

Create a new/empty file of your own under the `lab2` folder named `Lab02_Q2.java` with a class with the same name. Write a Java program that simulates a "Solar System Weight Calculator". The program should input the **masses (kg)** of two different space probes (Probe 1 and Probe 2) and calculate and display their corresponding weights (in Newtons) on **Earth, Moon, Mars, and Jupiter**.

- Earth:  $9.807 \text{ m/s}^2$
- Moon:  $1.620 \text{ m/s}^2$
- Mars:  $3.711 \text{ m/s}^2$
- Jupiter:  $24.790 \text{ m/s}^2$

### Sample Run:

```
Enter the mass of the first probe (kg): 100
Enter the mass of the second probe (kg): 55.5
          EARTH      MOON      MARS      JUPITER
PROBE ONE (100.0kg)  980.7      162.0      371.1      2479.0
PROBE TWO (55.5kg)   544.3      89.9       206.0      1375.8
```

## 3. Working with Strings

Create a new/empty file of your own under the `lab2` folder named `Lab02_Q3.java` with a class with the same name. Write a Java program that inputs a library book entry in a specific raw format. Using String methods, your program should output the book details in a readable sentence.

**Input Format:** Title {Author} [Year] \_ Publisher

### Assumptions:

- The symbols `{`, `}`, `[`, `]`, and `_` are always present in this order.
- There may be extra spaces around the words and symbols. You must remove them using `trim()`.

### Sample Run:

```
Enter book information: The Great Gatsby { F. Scott Fitzgerald } [ 1925 ] _
Scribner
The book "The Great Gatsby" was written by F. Scott Fitzgerald in 1925 and
published by Scribner.
```